

capture high-speed waveforms. It has a detachable panel computer too.

Large memory capacity, high accuracy, synchronous sampling and transmission, and above all a smart wireless control system are the unique features of light flicker analysers.

Ageing and life-cycle tester

As per standards, the life-cycle test is for up to 50,000 hours, so we can't mention it as minimum timeline. LED luminaire manufacturers generally place their products in the life-test rack.

Ageing tester is another important equipment to measure photometric and electrical parameters of LED luminaires to check their capability to withstand field usage.

But both these equipment are large, requiring a lot of space. So test solution manufacturers have released a product that combines functions of both the life tester and the ageing test machine. Built as per IESNA LM-82-12 and IESNA LM-80-08 standards



Fig. 5: Accelerated ageing and life testing equipment (Image courtesy: www.everfine.net)

for LED luminaires and LED/LED arrays, respectively, it can be used for accelerated ageing and lumen maintenance measurement, lifetime evaluation, temperature tests and chromaticity shift measurement of LED products.

Besides, manufacturers are coming up with portable biological test equipment for LED luminaires. These equipment integrate spectroradiometer as well as spectra luminous meter.

Challenges

Most of the LED luminaire test equipment are costly as these are imported from manufacturers based out of India. In such a situation, SMEs resort to third-party testing from laboratories like Bharat Test House, UL, Yadav Test Lab and ERDL. These labs provide LED manufacturers with the facility to test their LED luminaires. Costs are high for each test and, in case of failure, usually, the customer has to pay for another cycle of test.

As LED luminaire manufacturers desist from investing money in test equipment, they are unable to master their in-house products. However, there are a few major players like Osram, Philips, Surya, Goldwyn and Havells who have their own NABL-certified full-fledged laboratories and are able to offer quality products. An upgraded test equipment for LEDs and LED luminaires is a one-time investment that proves advantageous in the long run. **EFY**

support@efy.in

Do you have a query, suggestion or a complaint?

**You can e-mail it to
support@efy.in
and we will take care
of the rest.**



electronics
FOR YOU
www.electronicshobby.com

ELECTRONICS
BAZAAR
www.ebayindia.com

OpenSource
For All
www.opensourceforu.com

EFY GROUP
Technology Drives Us
www.efy.in